Musanze Harvest Intelligence Service

Technical report - SWE 3513 Artificial Intelligence, Individual Assignment 2 - verification code **AI-A2-6972**

1. Identity and decision problem

Field	Value
Full name	Haroun Mohammed Arbab Yousef
Registration number	25/26972
Institutional username	ug2526972@ines.ac.rw
Verification code	AI-A2-6972
Random seed (LLLL)	6972
Base rows	242 (240 + 6972 mod 41)

The stakeholder is the field officer of a potato cooperative in Musanze. Farm records arrive in inconsistent spreadsheets, internet access is unreliable and the officer needs local, evidence-based support. The service answers three questions: (1) how many tons will a farm harvest, (2) is the farm at high production risk, and (3) which farms share similar growing conditions?

Ethical boundary. This is a classroom decision-support prototype built on synthetic data. It must not replace an agronomist and must never be used to decide access to finance, insurance or public services. Every automated answer requires human review by the field officer.

2. Data and reproducibility

The dataset is generated by `data/generate_data.py` with `numpy.random.default_rng(6972)`. The seed is the last four digits of the registration number (LLLL = 6972). The generator writes 242 base rows, sets 12 soil moisture values and 12 fertilizer values to missing with the same generator stream, then appends 3 exact duplicate rows. All numeric values use four decimal places.

Column	Rule	Role
rainfall_mm	Uniform 600-1400	feature
avg_temp_c	Uniform 12-26	feature
soil_moisture_pct	Uniform 20-80 (12 missing)	feature
fertilizer_kg	Uniform 0-250 (12 missing)	feature
field_size_ha	Uniform 0.2-4.0	feature
season	A or B	categorical feature
cooperative_support	0 or 1	binary feature
yield_tons	given linear formula + N(0, 0.80), floor 0.50	regression target
high_risk	yield per hectare below median	classification target

Evidence	Value
Rows in CSV	245
Rows after cleaning duplicates	242
Missing values detected	{"soil_moisture_pct": 12, "fertilizer_kg": 12}
Exact duplicates detected	3
SHA-256 of final CSV	a05f4763ec64a6967d3bd49a679c6ae351d143bb1ffe5827fe99e685a66d4474

Fingerprint file record: sha256: a05f4763ec64a6967d3bd49a679c6ae351d143bb1ffe5827fe99e685a66d4474

3. Pipeline design

Split. Exact duplicate rows are removed first, then a single fixed stratified split (random_state = 6972, stratify on high_risk) produces 193 training rows and 49 held-out test rows. The test rows are never used for fitting.

Leakage control. Numeric missing values are imputed with the training median; numeric features are standardised; `season` is one-hot encoded with unknown-category tolerance. All imputers, the scaler and the encoder live inside one ColumnTransformer and are fitted on the training split only, inside each model Pipeline. yield_tons, yield per hectare and high_risk are excluded from every feature set, and clustering never reads a target column.

Candidates. Regression: mean baseline, LinearRegression, RandomForestRegressor(200). Classification: most-frequent baseline, LogisticRegression, RandomForestClassifier(200). Clustering: KMeans with K = 2..6 on standardised input features.

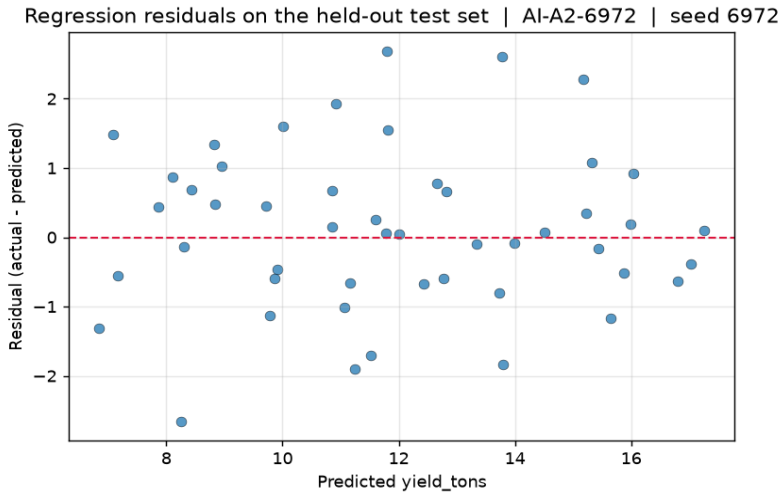
4. Regression and classification

Regression (target yield_tons)

Model	MAE (t)	RMSE (t)	R2
mean_baseline	2.6024	3.1208	-0.0005
linear_regression	0.895	1.1486	0.8645
random_forest	1.1654	1.4792	0.7752

Selected model: **linear_regression**. Selected by lowest test MAE among all evaluated models: linear_regression MAE=0.895 vs random_forest MAE=1.1654 (baseline MAE=2.6024). Test set has 49 rows, split with random_state=6972.

Error interpretation. On the test set the largest absolute error is row 210: actual 14.4816 t versus predicted 11.7927 t (error 2.6889 t). The model under-predicts this farm. The residual plot shows errors are centred on zero with roughly constant spread, which is what we expect when the target is generated by a linear formula with Gaussian noise; the largest errors are single noisy draws rather than a systematic bias. RandomForest loses to the linear model because trees cannot extrapolate the continuous linear trend and split it into steps.



Classification (target high_risk)

Model	Precision	Recall	F1	ROC-AUC
majority_baseline	0.0	0.0	0.0	0.5
logistic_regression	0.9259	1.0	0.9615	0.995
random_forest	0.9565	0.88	0.9167	0.9867

Selected model: **logistic_regression**. Confusion matrix [[TN, FP], [FN, TP]] = [[22, 2], [0, 25]]: 2 false positive(s) and 0 false negative(s). ROC-AUC 0.995 shows the ranking is almost perfect, while the F1 shortfall comes from borderline farms near the median yield-per-hectare where the random noise flips the label; this is label noise in the generator, not a modelling mistake.

Error interpretation. Row 74 is classified 1 with probability 0.5069 but the actual label is 0. Its field size makes the yield-per-hectare ratio sit just beside the median, so a small residual pushes the label across the threshold. Consequence for the officer: borderline farms should be inspected in person rather than trusted to the model.

Selected by highest test F1 (tie-break ROC-AUC): logistic_regression F1=0.9615, ROC-AUC=0.995 vs baseline F1=0.0, ROC-AUC=0.5.

5. Clustering results

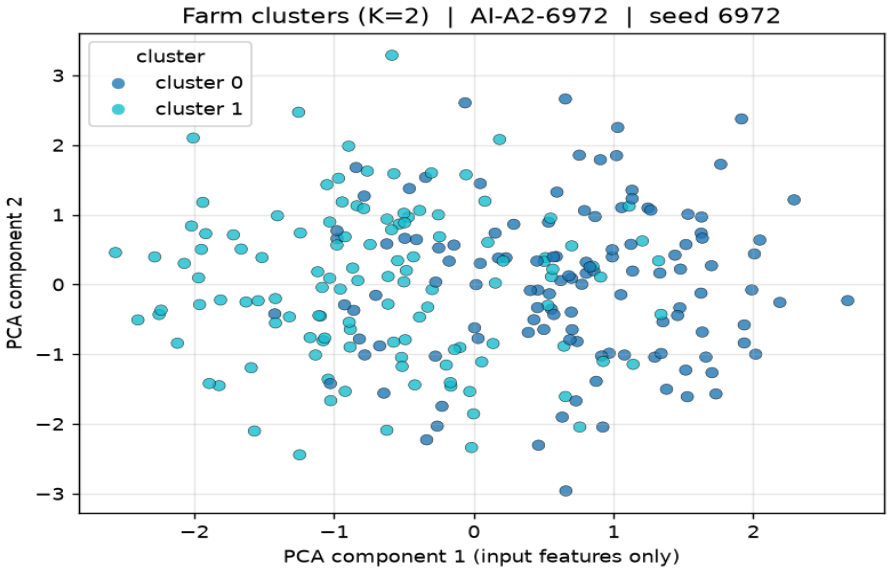
K	Silhouette
2	0.1573
3	0.1348
4	0.1431
5	0.1395
6	0.1396

Selected K = **2** by highest silhouette score. K=2 gives the highest silhouette score on the training set (0.1573) among K=2..6. Silhouette ranges from -1 to 1; higher means better separated, denser clusters.

Cluster 0 (121 of 242 farms; dominant season A): Cluster 0 (A-leaning farms) has; the highest rainfall_mm (1015.53), soil_moisture_pct (52.55), fertilizer_kg (137.38), field_size_ha (2.15) among all clusters; the lowest avg_temp_c (19.12), cooperative_support (0.00) among all clusters; its strongest difference is avg_temp_c=19.12 (about 0.7 standard deviations from the overall mean).

Cluster 1 (121 of 242 farms; dominant season A): Cluster 1 (A-leaning farms) has; the highest avg_temp_c (19.13), cooperative_support (1.00) among all clusters; the lowest rainfall_mm (994.67), soil_moisture_pct (48.59), fertilizer_kg (124.99), field_size_ha (2.05) among all clusters; its strongest difference is rainfall_mm=994.67 (about 0.7 standard deviations from the overall mean).

Operational meaning and honest caveat. The silhouette value is modest and the centres differ mainly in cooperative support, while rainfall, temperature, moisture, fertilizer and field size are almost identical between clusters. The split is therefore a weak segmentation: it separates farms that received cooperative support from those that did not, not genuinely distinct soil or climate zones. The officer may use clusters for exploratory grouping only, and this result should be re-checked with real farm data that has more varied structure before any advice is based on it.



6. API and testing

Endpoint	Request	Response	Failure behaviour
GET /health	none	status, verification code, seed, model_loaded	
GET /model-info	none	identity, features, metrics, intended use	
POST /predict	JSON FarmRecord (7 typed fields)	yield_tons, high_risk, risk_probability, cluster, verification code	422 with field-level message for missing, wrong-ty

Valid request/response (examples/valid.json):

{"rainfall_mm": 950.0, "avg_temp_c": 18.5, "soil_moisture_pct": 55.0, "fertilizer_kg": 120.0, "field_size_ha": 1.8, "season": "A", "cooperative_support": 1} -> 200 OK, yield 12.0151 t, high_risk 0, probability 0.2241, cluster 1

Invalid request (examples/invalid.json):

rainfall_mm 1500, avg_temp_c 40, soil_moisture_pct 95, fertilizer_kg -5, field_size_ha 9.9, season C, cooperative_support 7 -> 422 invalid input with one message per broken rule

Test summary. The pytest suite exercises failure paths, not only happy paths: invalid seed rejection, student-id guard, injected-defect detection, range violation detection, dataset reproducibility (identical SHA-256 on repeated generation), CLI valid and invalid runs, missing field, wrong type, out-of-range value, unknown field and the FastAPI /health, /model-info and /predict endpoints.

7. Model card and reflection

Item	Statement
Intended use	Estimate harvest tons, flag high production risk and group similar farms for a Musanze potato cooperative field offic
Data status	100% synthetic data generated from the student's seed. No real farm was measured; no personal data is processe
Limitations	Linear generator makes linear_regression hard to beat; clustering is weakly separated; missing values are imputed
Risks	Over-trust in a prototype; risk labels could be read as credit or insurance signals; distribution shift on real farms.
Prohibited use	Determining access to finance, insurance or public services; replacing an agronomist; any use on real farms withou
Human oversight	Every prediction is a recommendation for review by the field officer, who remains accountable for the decision.
What to improve	Real, multi-season cooperative records; a held-out year for temporal validation; calibrated probabilities; cost-sensiti

All figures in this report were read directly from the generated artifacts: regression_metrics.json, classification_metrics.json, clustering_report.json, test_predictions.csv and data_fingerprint.txt, so the report and the code agree by construction. Dataset SHA-256: a05f4763ec64a6967d3bd49a679c6ae351d143bb1ffe5827fe99e685a66d4474. Verification code: AI-A2-6972.